

DATA ANALYTICS

Interview Preparation — 60 Technical Questions & Answers

Candidate: Nishant Anand

Role Applied For: Data Analyst / Data Analytics

Target Package: 12+ LPA

Resume

NISHANT ANAND

Linkedin: [linkedin.com/in/nishantanand-571b58205](https://www.linkedin.com/in/nishantanand-571b58205)
Github: <https://github.com/nishantanand>

Email: nishantanand2021@gmail.com
Mobile: +91 8303445502

SKILLS

Languages:	Python ,C++,SQL
Web Dev:	HTML , CSS
Data & AI:	Power BI,TensorFlow, Keras, ETL, Informatica (IDQ)
Testing & QA:	Manual Testing
Tools & Databases:	Git, MS SQL Server, MongoDB, MS Office
Soft Skills:	Problem-Solving Skills, Leadership, Project Management, Adaptability

WORK EXPERIENCE

Data & Operations Specialist Power BI, ETL, SQL	Apr'23 – Apr'25
ConveGenius Edu Solutions Pvt. Ltd. Chandauli, India	
- Significantly improved learning outcomes across Chandauli District by identifying data gaps and sharing actionable insights with key stakeholders. - Analyzed large operational datasets using Python & Excel; authored weekly KPI reports and interactive Power BI dashboards across 20+ government schools, reducing decision turnaround by ~30%. - Designed & implemented structured ETL workflows consolidating multi-source field data into a centralised SQL Server database, cutting manual aggregation effort by ~40%. - Built drill-down Power BI dashboards (grade, school, subject-level) for real-time programme monitoring; streamlined reporting templates, saving ~5 manual hours/week.	

PROJECTS

Electoral Voter Data Analysis & ML Prediction Python, Pandas, Scikit-Learn, EDA	Mar'26 – Apr'26
- Processed & analyzed 341,674 real-world voter records across 420 booths; applied IQR methods for data cleaning and conducted comprehensive EDA on demographics. - Engineered predictive features from regional text fields, training 5 ML models and boosting baseline accuracy from ~59% to ~94% for voter gender prediction.	
India General Election Results 2024 Analysis Power BI, Power Query, Advanced DAX	Mar'26 – Apr'26
- Designed a 5-page interactive Power BI report with a hybrid star schema data model authored advanced DAX measures to compute vote shares dynamically with drill-through navigation. - Created advanced DAX measures and implemented slicers, drill-through, and navigation features.	
BlinkIt Sales & Outlet Performance Dashboard Excel, Power Query	Sep'25 – Nov'25
- Built an interactive Excel dashboard analyzing 8,500+ records; implemented ETL via Power Query to track sales performance, outlet efficiency, and customer behavior (Total Sales: ₹1.20M). - Designed key KPIs (₹1.20M Total Sales, Avg Sales, Item Count, Ratings) and implemented ETL using Power Query for clean, consolidated data. - Derived actionable insights (Tier-3 cities highest sales, 65% low-fat product demand, peak time trends) to support business decisions.	

CERTIFICATES

Cisco Data Analytics Essentials	Feb'25
IBM SkillsBuild – AI Fundamentals	Aug'21
IBM Open P-Tech – Cyber Security Certificate	Jun '21

ACHIEVEMENTS

- "North Star" Recognition – ConveGenius: Recognized as the "North Star" for exceptional dedication, leadership & innovation over 2 years; officially cited by the company as a role model.
- Sports Excellence: Represented college at Zonal level sports competition in 2nd year, demonstrating teamwork and competitive spirit.
- School Leadership: Served as Head Boy (Class 11) and Yellow House Captain (Class 10) while maintaining outstanding academic records throughout.

EDUCATION

Lovely Professional University	Aug'25 – Present
Masters of Technology in Data Science And Analytics CGPA:-8.29	Phagwara, Punjab
Ashoka Institute of Technology & Management	Jun 2019 – Jul 2023
B.Tech – Computer Science & Engineering 77.62% Dr. APJ Abdul Kalam Technical University Varanasi, India	
RN Public School (12th – Intermediate)	2018 – 2019
	Rohania, Varanasi
BPS Public School (10th – High School)	2016 – 2017
CGPA: 9.5	Naubatpur, Varanasi

SQL

Q1. What is the difference between WHERE and HAVING clauses? [SQL]

Ans: WHERE filters individual rows before grouping/aggregation, while HAVING filters groups after GROUP BY is applied. WHERE cannot use aggregate functions, but HAVING can.

Q2. What is the difference between INNER JOIN and LEFT JOIN? [SQL]

Ans: INNER JOIN returns only matching rows from both tables, whereas LEFT JOIN returns all rows from the left table plus matched rows from the right table (NULL if no match).

Q3. What are Primary Key and Foreign Key? [SQL]

Ans: A Primary Key uniquely identifies each record in a table and cannot be NULL. A Foreign Key is a column referencing the Primary Key of another table, enforcing referential integrity.

Q4. What is the difference between DELETE, TRUNCATE, and DROP? [SQL]

Ans: DELETE removes rows with optional conditions and can be rolled back; TRUNCATE removes all rows instantly and resets identity, with minimal logging; DROP removes the entire table structure permanently.

Q5. What are Window Functions? Give an example. [SQL]

Ans: Window functions perform calculations across a set of rows related to the current row without collapsing them, e.g. ROW_NUMBER(), RANK(), and SUM() OVER(PARTITION BY ...) used for running totals or rankings.

Q6. What is a Common Table Expression (CTE)? [SQL]

Ans: A CTE, defined using WITH, is a temporary named result set that improves query readability and can be referenced within a SELECT, INSERT, UPDATE, or DELETE statement.

Q7. What is the difference between UNION and UNION ALL? [SQL]

Ans: UNION combines results of two queries and removes duplicate rows, while UNION ALL combines results without removing duplicates, making it faster.

Q8. What is normalization and why is it used? [SQL]

Ans: Normalization organizes data to reduce redundancy and improve integrity by dividing large tables into smaller related tables using keys, typically up to 3NF in analytics work.

Q9. How do you find duplicate records in a table using SQL? [SQL]

Ans: Use GROUP BY on the relevant columns with HAVING COUNT(*) > 1, or use ROW_NUMBER() OVER(PARTITION BY column) and filter rows where the row number is greater than 1.

Q10. What is the difference between clustered and non-clustered index? [SQL]

Ans: A clustered index physically sorts and stores table rows in order of the indexed column (one per table), while a non-clustered index creates a separate structure with pointers to the actual rows (multiple allowed).

Q11. What is a subquery and where can it be used? [SQL]

Ans: A subquery is a query nested inside another SQL statement; it can be used in SELECT, WHERE, FROM, or HAVING clauses to filter or compute intermediate results.

Q12. How would you calculate a running total in SQL Server? [SQL]

Ans: Use SUM(amount) OVER (ORDER BY date) or SUM(amount) OVER (PARTITION BY category ORDER BY date) to compute a cumulative total row by row.

Power BI & DAX

Q13. What is the difference between Power Query and DAX in Power BI? [Power BI & DAX]

Ans: Power Query is used for ETL — importing, cleaning, and shaping data before it loads into the model, while DAX (Data Analysis Expressions) is used to create calculated columns and measures for analysis after loading.

Q14. What is the difference between a Calculated Column and a Measure in DAX? [Power BI & DAX]

Ans: A calculated column is computed row-by-row and stored in the table, increasing model size, whereas a measure is computed dynamically at query time based on the current filter context, making it more memory efficient.

Q15. What is a Star Schema and why is it preferred in Power BI? [Power BI & DAX]

Ans: A star schema has a central fact table connected to surrounding dimension tables, simplifying relationships and improving DAX performance compared to a flat or snowflake structure.

Q16. Explain CALCULATE() function in DAX. [Power BI & DAX]

Ans: CALCULATE() evaluates an expression in a modified filter context, allowing you to override or add filters, e.g. CALCULATE(SUM(Sales), Region="North") to compute sales only for the North region.

Q17. What is the difference between Filter Context and Row Context in DAX? [Power BI & DAX]

Ans: Row context evaluates an expression row by row, typically in calculated columns or iterator functions, while filter context is the set of filters (from slicers, rows, columns) applied when a measure is evaluated.

Q18. What is a drill-through in Power BI? [Power BI & DAX]

Ans: Drill-through lets users right-click a data point on one report page to navigate to a detailed page filtered to that specific context, useful for exploring granular data like a single school or outlet.

Q19. What is the difference between Power BI Import mode and DirectQuery mode? [Power BI & DAX]

Ans: Import mode loads and compresses data into Power BI's in-memory model for fast performance, while DirectQuery sends live queries to the source each time, giving real-time data at the cost of speed.

Q20. What are slicers used for in Power BI dashboards? [Power BI & DAX]

Ans: Slicers are visual filter controls that let end users interactively filter report data by fields such as region, date, or category without editing the underlying report.

Q21. How do you optimize a slow Power BI report? [Power BI & DAX]

Ans: Reduce cardinality of columns, use star schema instead of flat tables, avoid excessive calculated columns, disable auto date/time, and use variables in DAX measures to avoid repeated evaluation.

Q22. What is the ALL() function used for in DAX? [Power BI & DAX]

Ans: ALL() removes filters from a specified table or column, which is useful for calculating totals unaffected by slicers, such as percentage of grand total.

Q23. What is Power Query M language used for? [Power BI & DAX]

Ans: M is the functional language behind Power Query used to define data transformation steps such as merging, appending, pivoting, and cleaning data during the ETL process.

Q24. How do you create a dynamic year-over-year (YoY) growth measure in DAX? [Power BI & DAX]

Ans: Use a measure like `YoY = DIVIDE(SUM(Sales[Amount]) - CALCULATE(SUM(Sales[Amount]), SAMEPERIODLASTYEAR(Dates[Date])), CALCULATE(SUM(Sales[Amount]), SAMEPERIODLASTYEAR(Dates[Date])))` to compare current and prior year performance.

Excel & Power Query

Q25. What is the difference between VLOOKUP and INDEX-MATCH? [Excel & Power Query]

Ans: VLOOKUP searches a value in the leftmost column and returns data from a column to the right, but only searches left-to-right, while INDEX-MATCH is more flexible, faster on large data, and can look in any direction.

Q26. What are Pivot Tables used for in Excel? [Excel & Power Query]

Ans: Pivot Tables summarize and aggregate large datasets quickly, allowing grouping, filtering, and cross-tabulation of data such as sales by outlet and month without writing formulas.

Q27. What is Power Query in Excel used for? [Excel & Power Query]

Ans: Power Query is Excel's ETL tool used to import, clean, merge, and transform data from multiple sources before loading it into a worksheet or the data model.

Q28. How do you remove duplicate records in Excel? [Excel & Power Query]

Ans: Select the data range and use Data > Remove Duplicates, or use a formula like COUNTIF to flag duplicates before filtering them out.

Q29. What is the difference between COUNTIF and SUMIF? [Excel & Power Query]

Ans: COUNTIF counts the number of cells meeting a condition, while SUMIF sums the values of cells meeting a condition, both operating on a single criteria range.

Q30. What are conditional formatting and its use in dashboards? [Excel & Power Query]

Ans: Conditional formatting visually highlights cells based on rules, such as color-scaling sales figures or flagging outlets below target, making trends and outliers easy to spot.

Q31. How do you handle circular references in Excel? [Excel & Power Query]

Ans: Identify the cell causing the loop using Formulas > Error Checking > Circular References, then redesign the formula logic or enable iterative calculation only if intentional, such as for goal-seek models.

Q32. What is the purpose of the XLOOKUP function? [Excel & Power Query]

Ans: XLOOKUP is a modern replacement for VLOOKUP/HLOOKUP that searches in any direction, supports exact match by default, and handles missing values gracefully with an if-not-found argument.

Q33. How would you create a KPI dashboard in Excel for outlet performance? [Excel & Power Query]

Ans: Use Power Query to clean and consolidate raw sales data, build Pivot Tables for KPIs like total sales and average ticket size, then visualize them with charts and slicers on a dashboard sheet.

Q34. What is the difference between a Pivot Table and Power Pivot? [Excel & Power Query]

Ans: A standard Pivot Table summarizes one flat table, while Power Pivot uses the Data Model to build relationships across multiple tables and supports DAX measures for more advanced analysis.

Python & Pandas

Q35. What is the difference between a Python list and a NumPy array? [Python & Pandas]

Ans: A list can hold mixed data types and is slower for numerical operations, while a NumPy array requires uniform data types and supports fast vectorized mathematical operations.

Q36. How do you handle missing values in a Pandas DataFrame? [Python & Pandas]

Ans: Use `df.isnull().sum()` to detect missing values, then handle them using `df.dropna()` to remove or `df.fillna()` with mean, median, or a constant depending on the data context.

Q37. What is the difference between `loc[]` and `iloc[]` in Pandas? [Python & Pandas]

Ans: `loc[]` selects data by label/index name, while `iloc[]` selects data by integer position, regardless of the actual index labels.

Q38. How do you merge two DataFrames in Pandas? [Python & Pandas]

Ans: Use `pd.merge(df1, df2, on='key_column', how='inner'/'left'/'right'/'outer')` to combine DataFrames similar to SQL joins based on a common key.

Q39. What is the IQR method and how is it used for outlier detection? [Python & Pandas]

Ans: IQR is the range between the 25th and 75th percentile ($Q3 - Q1$); values below $Q1 - 1.5 * IQR$ or above $Q3 + 1.5 * IQR$ are flagged as outliers and can be capped or removed.

Q40. What is the difference between `groupby()` and `pivot_table()` in Pandas? [Python & Pandas]

Ans: `groupby()` aggregates data based on one or more keys returning a grouped object, while `pivot_table()` reshapes data into a matrix format with rows, columns, and aggregated values, similar to Excel pivot tables.

Q41. What is Exploratory Data Analysis (EDA) and why is it important? [Python & Pandas]

Ans: EDA is the process of summarizing, visualizing, and understanding a dataset's structure, distributions, and relationships before modeling, helping to detect data quality issues and inform feature engineering.

Q42. How do you encode categorical variables for a machine learning model? [Python & Pandas]

Ans: Use one-hot encoding (`pd.get_dummies`) for nominal categories with few unique values, or label encoding for ordinal categories, choosing based on the model type and cardinality.

Q43. What is feature engineering? Give an example from voter data analysis. [Python & Pandas]

Ans: Feature engineering creates new predictive variables from raw data, such as deriving a 'region' feature from a voter's address text field to improve gender prediction accuracy.

Q44. What is the difference between `apply()`, `map()`, and `applymap()` in Pandas? [Python & Pandas]

Ans: `map()` works element-wise on a Series, `apply()` works on a Series or DataFrame along an axis (row/column) and can use complex functions, while `applymap()` applies a function element-wise to an entire DataFrame.

Q45. How do you check and treat class imbalance in a classification dataset? [Python & Pandas]

Ans: Check with `value_counts()` on the target column; treat it using oversampling (SMOTE), undersampling, or class-weight adjustment in the model to prevent bias toward the majority class.

Q46. What is the difference between a Series and a DataFrame in Pandas? [Python & Pandas]

Ans: A Series is a one-dimensional labeled array holding a single column of data, while a DataFrame is a two-dimensional labeled structure holding multiple Series (columns) together.

ETL & Informatica IDQ

Q47. What is ETL and why is it important in data analytics? [ETL & Informatica IDQ]

Ans: ETL stands for Extract, Transform, Load — it consolidates data from multiple sources into a clean, structured format in a central database, enabling reliable reporting and analysis.

Q48. What is Informatica IDQ used for? [ETL & Informatica IDQ]

Ans: Informatica Data Quality (IDQ) is used to profile, cleanse, standardize, and validate data, ensuring accuracy and consistency before it is loaded into a data warehouse.

Q49. What is data profiling and why is it done before ETL? [ETL & Informatica IDQ]

Ans: Data profiling examines source data to understand its structure, completeness, and quality issues like nulls or duplicates, helping design accurate transformation rules before loading.

Q50. What is the difference between a Data Warehouse and a Database? [ETL & Informatica IDQ]

Ans: A database is optimized for transactional operations (OLTP) with frequent reads/writes, while a data warehouse is optimized for analytical queries (OLAP), storing large historical, aggregated data for reporting.

Q51. What is data standardization in ETL workflows? [ETL & Informatica IDQ]

Ans: Data standardization converts data into a consistent format, such as unifying date formats or naming conventions across multiple source systems, before consolidation into a central database.

Q52. How would you design an ETL workflow to consolidate multi-source school data into SQL Server? [ETL & Informatica IDQ]

Ans: Extract raw files from each source, use staging tables to clean and validate data (removing duplicates/nulls), apply transformation rules to standardize formats, and load the final data into central SQL Server tables using scheduled jobs.

Statistics & Machine Learning

Q53. What is the difference between correlation and causation? [Statistics & Machine Learning]

Ans: Correlation indicates two variables move together, while causation means one variable directly causes a change in another; correlation alone does not prove causation.

Q54. What is the difference between Mean, Median, and Mode? [Statistics & Machine Learning]

Ans: Mean is the average of all values, Median is the middle value when sorted, and Mode is the most frequently occurring value; median is preferred for skewed data with outliers.

Q55. What is overfitting in machine learning and how do you prevent it? [Statistics & Machine Learning]

Ans: Overfitting occurs when a model learns noise in training data and performs poorly on new data; it can be prevented using cross-validation, regularization, simpler models, or more training data.

Q56. How do you evaluate a classification model's performance? [Statistics & Machine Learning]

Ans: Use metrics like accuracy, precision, recall, F1-score, and a confusion matrix; accuracy alone can be misleading on imbalanced datasets like the voter gender prediction task.

Q57. What is the difference between Type I and Type II errors? [Statistics & Machine Learning]

Ans: A Type I error is rejecting a true null hypothesis (false positive), while a Type II error is failing to reject a false null hypothesis (false negative).

Q58. What is Standard Deviation and why is it useful in analytics? *[Statistics & Machine Learning]*

Ans: Standard deviation measures how spread out data values are from the mean; a low value indicates data points cluster near the mean, useful for understanding consistency in KPIs.

Q59. Which 5 ML models are commonly compared for a classification task like gender prediction, and how do you pick the best one? *[Statistics & Machine Learning]*

Ans: Common models include Logistic Regression, Decision Tree, Random Forest, KNN, and Naive Bayes; the best one is chosen by comparing accuracy, precision/recall, and cross-validation performance on the test set.

Q60. What is the difference between supervised and unsupervised learning? *[Statistics & Machine Learning]*

Ans: Supervised learning trains a model on labeled data to predict outcomes (e.g. gender prediction), while unsupervised learning finds hidden patterns or groupings in unlabeled data (e.g. clustering).